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Lipschitz stability in inverse parabolic problems by the Carleman estimate

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Abstract. We consider a system

$$y_t(t,x) = -\mathcal{A}y(t,x) + g(t,x) \qquad 0 < t < T, x \in \Omega$$
$$y(\theta,x) = y_0(x) \qquad x \in \Omega$$

with a suitable boundary condition, where $\Omega \subset \mathbb{R}^n$ is a bounded domain, $-\mathcal{A}$ is a uniformly elliptic operator of the second order whose coefficients are suitably regular for (t,x), $\theta \in]0,T[$ is fixed, and a function $g(t,x)$ satisfies

$$|g_t(t,x)| \leqslant C|g(\theta,x)| \qquad \text{for } (t,x) \in [0,T] \times \overline{\Omega}.$$

Our inverse problems are determinations of g using overdetermining data $y_{|]0,T[\times\omega}$ or $\{y_{|]0,T[\times\Gamma_0}, \nabla y_{|]0,T[\times\Gamma_0}\}$, where $\omega \subset \Omega$ and $\Gamma_0 \subset \partial\Omega$. Our main result is the Lipschitz stability in these inverse problems. We also consider the determination of $f = f(x)$, $x \in \Omega$ in the case of $g(t,x) = f(x)R(t,x)$ with given R satisfying $R(\theta,\cdot) > 0$ on $\overline{\Omega}$. Finally, we discuss an upper estimation of our overdetermining data by means of f.

1. Introduction

We consider

$$\begin{aligned} &y_t(t,x) = -\mathcal{A}y(t,x) + g(t,x) && 0 < t < T, x \in \Omega \\ &l_1(x)\frac{\partial y}{\partial \nu_{\mathcal{A}}}(t,x) + l_2(x)y(t,x) = 0 && 0 < t < T, x \in \partial\Omega \\ &y(\theta,x) = y_0(x) && x \in \Omega \end{aligned} \tag{1.1}$$

where

$$-\mathcal{A}y(t,x) = \sum_{i,j=1}^{n} \frac{\partial}{\partial x_i}\left(a_{ij}(t,x)\frac{\partial y}{\partial x_j}(t,x)\right) - \sum_{i=1}^{n} b_i(t,x)\frac{\partial y}{\partial x_i}(t,x) - c(t,x)y(t,x)$$

is a uniformly elliptic operator of second order whose coefficients are suitably regular in $(t,x) \in [0,T] \times \overline{\Omega}$, $\Omega \subset \mathbb{R}^n$ is a bounded domain with boundary $\partial\Omega$ of class C^2, l_1 and l_2 are given on $\partial\Omega$ suitably, $\theta \in]0,T[$ is fixed. Moreover, the function $g(t,x)$ satisfies

$$\left|\frac{\partial g(t,x)}{\partial t}\right| \leqslant C|g(\theta,x)| \qquad \text{for almost all } (t,x) \in [0,T] \times \overline{\Omega} \tag{1.2}$$

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for some constant $C > 0$. Henceforth we set $Q =]0, T[\times\Omega$.

Our first inverse problem consists of determining $g(t, x)$ by using overdetermining data $y_{|]0,T[\times\omega}$ where $\omega \subset \Omega$ is an arbitrary sub-domain.

Simultaneously with problem (1.1) we consider a similar problem

$$\begin{aligned} &y_t(t,x) = -\mathcal{A}y(t,x) + g(t,x) \qquad 0 < t < T, x \in \Omega \\ &\left(l_1(x)\frac{\partial y}{\partial \nu_{\mathcal{A}}} + l_2(x)y\right)_{|\partial\Omega\setminus\Gamma_0} = 0 \qquad 0 < t < T \\ &y(\theta, x) = y_0(x) \qquad x \in \Omega \end{aligned} \tag{1.3}$$

where Γ_0 is an arbitrarily fixed relatively open subset of $\partial\Omega$. For (1.3) we consider an inverse problem of determining $g = g(t, x)$ from overdetermining data $y_{|]0,T[\times\Gamma_0}$ and $\nabla y_{|]0,T[\times\Gamma_0}$.

We mention one special but very important case: an unknown right-hand side of a parabolic equation has the form $g(t, x) = R(t, x)f(x)$ where $R(t, x)$ is given and satisfies

$$|R(\theta, x)| > 0 \qquad x \in \overline{\Omega},\ \ R \in C^{1,0}(\overline{Q}) \tag{1.4}$$

while f is an unknown function. Hereafter we set

$$C^{1,0}(\overline{Q}) = \left\{ y(t,x) | y, \frac{\partial y}{\partial t} \in C(\overline{Q}) \right\}.$$

One can easily check that under condition (1.4), $R(t, x)f(x)$ of the right-hand side satisfies (1.2) for arbitrary f.

In this case we consider two types of inverse problems.

In the first inverse problem we determine $f = f(x)$, $x \in \Omega$ from overdetermining data $y_{|]0,T[\times\omega}$ in (1.5):

$$\begin{aligned} &y_t(t,x) = -\mathcal{A}y(t,x) + f(x)R(t,x) \qquad 0 < t < T, x \in \Omega \\ &l_1(x)\frac{\partial y}{\partial \nu_{\mathcal{A}}}(t,x) + l_2(x)y(t,x) = 0 \qquad 0 < t < T, x \in \partial\Omega \\ &y(\theta, x) = y_0(x) \qquad x \in \Omega. \end{aligned} \tag{1.5}$$

In the second, we determine $f = f(x)$, $x \in \Omega$ from overdetermining data $y_{|]0,T[\times\Gamma_0}$ and $\nabla y_{|]0,T[\times\Gamma_0}$ in (1.6):

$$\begin{cases} y_t(t,x) = -\mathcal{A}y(t,x) + f(x)R(t,x) & 0 < t < T, x \in \Omega \\ \left(l_1(x)\dfrac{\partial y}{\partial \nu_{\mathcal{A}}} + l_2(x)y\right)|_{\partial\Omega\setminus\Gamma_0} = 0 & 0 < t < T \\ y(\theta, x) = y_0(x) & x \in \Omega. \end{cases} \tag{1.6}$$

The function f models a spatial distribution of source term and the determination of f is called an inverse source problem for a parabolic equation.

These inverse problems for (1.5) and (1.6) can also be considered as determination of a coefficient $p = p(x)$, $x \in \Omega$ in

$$\begin{aligned} &u_t(t,x) = -\mathcal{A}u(t,x) + p(x)u(t,x) \qquad 0 < t < T, x \in \Omega \\ &u(\theta, x) = u_0(x) \qquad x \in \Omega \end{aligned} \tag{1.7}$$

with suitable boundary conditions. In fact, subtraction of (1.7) with different $p(x)$ and $q(x)$ yields a parabolic equation in (1.5) or (1.6), and for the determination of p, q it suffices to discuss the system (1.5) or (1.6). On the basis of our key lemmata 2.2 and 2.4 in section 2, we can justify the argument, and in a forthcoming paper we will discuss details.

In this paper, we mainly take observations which can be restricted to an arbitrarily small sub-domain ω or an arbitrarily small sub-boundary Γ_0 along any time interval $]0, T[$. These kinds of observations seem minimal for the unique determination, and can be practically understood as limits of observation data when we can increase numbers of observation points.

We exclusively discuss the determination of f by a single measurement of overdetermining data, unlike by the Dirichlet-to-Neumann map (see, e.g. [11–13]). For the case of a single measurement we refer to the papers by the Carleman estimate: Bukhgeim and Klibanov [2], Klibanov [15], Isakov [10, 13], where conormal derivatives are taken as overdetermining data for determining f in (1.5). In Klibanov [15], in the case of $\theta = 0$, it is proved that

$$\frac{\partial y}{\partial \nu_{\mathcal{A}}}(t, x) \qquad x \in \partial\Omega, 0 < t < T : \text{ arbitrary}$$

uniquely determines f under the assumption that $|R(0, x)| > 0$, $x \in \overline{\Omega}$. The proof in [15] is based on the uniqueness in the corresponding inverse hyperbolic problem for $\theta = 0$, and the conormal derivative $\partial y/\partial \nu_{\mathcal{A}}$ is required on the whole boundary $\partial\Omega$. However, from the viewpoint of the unique continuation property for the parabolic equation (see, e.g. [18]), we can expect that f should be uniquely determined from the conormal derivative of y on any relatively open subset of $\partial\Omega$ along any short time interval. Isakov [10] gave a positive answer to this question for the case of $\theta > 0$, where he assumed boundary conditions only on any relatively open subset of $\partial\Omega$, and his uniqueness result is the best possible for the case of $\theta > 0$. However the uniqueness is an open problem for $\theta = 0$. The case of $\theta = 0$ is essentially difficult because for the application of the Carleman estimate, the parabolic equation must hold for t in a neighbourhood of θ, namely, for $\theta - \delta < t < \theta + \delta$ with some $\delta > 0$. In the case of $\theta = 0$, this requires extensions of solutions of the parabolic equation to $t < 0$, which is impossible in general. For inverse hyperbolic problems, the methodology by the Carleman estimate is similar and hyperbolic equations usually admit such extensions to $t < 0$. Thus, the uniqueness is solved for inverse hyperbolic problems also in the case of $\theta = 0$, provided that the overdetermining data, such as conormal derivatives, are given on a sufficiently large sub-boundary (e.g. on $\partial\Omega$). (See Isakov [10–13] and Klibanov [15].)

We shall next consider the stability for our inverse parabolic problem. For example, in (1.5), estimate the L^2-norm of f by means of a suitable norm of overdetermining data. The purpose of this paper is to establish the unconditional and global Lipschitz stability in the case of

$$0 < \theta < T.$$

The rest case of $\theta = T$ is discussed in a succeeding paper. By the unconditional and global estimate, we mean that the estimate holds in the whole domain and without assumptions on boundedness of norms of unknown functions. In view of the stability for the inverse parabolic problem in the multidimensional case, there are not many papers and, as for full-length papers, there are few. For conditional Hölder stability estimates, we can refer to Khaĭdarov [14], and a comment on p 577 of [15]. For the stability in other kinds of inverse parabolic problems, we can refer to [3, 4, 24].

Remark. For similar inverse problems for hyperbolic equations, a global and both-sided Lipschitz stability estimate is shown in [21], for example. A both-sided estimate guarantees the equivalence between norms of unknown functions and norms of overdetermining data. A method by a traditional Carleman estimate in [16] may prove such stability in our inverse

hyperbolic problems, although in [16], such inverse problems are not discussed. In contrast to inverse parabolic problems, it is easier to obtain stability estimates in inverse hyperbolic problems, because the hyperbolic equation can be solved both upwards and downwards in time, which enables us to take extensions of solutions to $t < 0$.

Our argument is based on a new Carleman estimate proved in [5, 7–9]. In comparison with [10, 13, 15], our Carleman estimate is advantageous in the following three points as far as we assume boundary conditions on the whole boundary.

(1) We can establish an unconditional and global Lipschitz stability estimate, which is stronger than conditional Hölder estimates which can be derived from [14, 15].

(2) Without any modification, we can treat the case where the coefficients of highest-order terms depend on t.

(3) We can reduce the regularity of y, g and R.

This paper employs a combination of the new Carleman estimate with the past idea of application of such estimates to inverse problems in [2, 10, 11, 13–15]. A technical advantage of the new Carleman estimate over the classical Carleman estimate is that it holds in the whole cylindrical domain $]0, T[\times\Omega$ (note that the classical one holds in sub-domains bounded by level sets). This advantage allows us to obtain the global Lipschitz stability.

This paper is composed of four sections. In section 2, we show the new Carleman estimate. In section 3 we prove our main results concerning the unconditional and global Lipschitz stability in our inverse problems. From the viewpoint of the parabolicity, the condition $\theta > 0$ is essential. If $\theta > 0$, then the 'forward' problem of determining $y_{|]0,T[\times\omega}$ in (1.1) or (1.5) with given R, y_0 and l_1, l_2, involves a backward parabolic equation and so the forward problem itself is ill-posed. Thus, we can expect some better stability in our problems which are inverse to such ill-posed problems. In section 4, we discuss ill-posedness and establish a conditional stability estimate in the forward problem for (1.5). Such conditional stability in section 4 balances with our Lipschitz stability in the inverse parabolic problem.

2. Carleman estimate

In this section, we show a Carleman estimate which holds over the whole domain Q, not only in level sets of some weight functions, for functions which vanish on $]0, T[\times\partial\Omega$. We need not assume that the functions under consideration have compact supports and so ours is different from the Carleman estimates presented in [10, 13, 15].

Let $(t, x) \in Q \equiv]0, T[\times\Omega$, where $\Omega \subset \mathbb{R}^n$ is a connected bounded domain with boundary $\partial\Omega$ of class C^2, $\nu = \nu(x) = (\nu_1(x), \ldots, \nu_n(x))$ the external unit normal vector to $\partial\Omega$ at x, $T \in]0, +\infty[$ is an arbitrary moment of time. Let ω be an arbitrary fixed sub-domain of Ω and $\Gamma_0 \subset \partial\Omega$ be an arbitrary fixed sub-domain of $\partial\Omega$ (in other words, Γ_0 is an arbitrary relatively open subset of $\partial\Omega$). Set $\Sigma =]0, T[\times\partial\Omega$ and $\Sigma_0 =]0, T[\times\Gamma_0$. In this paper we use the following function spaces:

$$W_2^k(\Omega) = \left\{ w(x) \,\middle|\, \|w\|_{W_2^k(\Omega)} \equiv \left(\sum_{|\beta|\leqslant k} \int_\Omega |D^\beta w|^2 \,\mathrm{d}x \right)^{\frac{1}{2}} < \infty \right\}$$

$$\beta = (\beta_1, \ldots, \beta_n) \in (\mathbb{Z}_+)^n \qquad |\beta| = \beta_1 + \cdots + \beta_n \qquad D_x^\beta = \frac{\partial^{\beta_1}}{\partial x_1^{\beta_1}} \cdots \frac{\partial^{\beta_n}}{\partial x_n^{\beta_n}}$$

$$W_2^{1,0}(Q) = \left\{ w(t,x) | w, \frac{\partial w}{\partial t}, \in L^2(Q) \right\}$$

$$W_2^{1,2}(Q) = \left\{ w(t,x) | w, \frac{\partial w}{\partial t}, \frac{\partial w}{\partial x_i}, \frac{\partial^2 w}{\partial x_i \partial x_j} \in L^2(Q), i, j = 1, \ldots, n \right\}$$

$$C^{1,1}(\overline{Q}) = \left\{ w(t,x) | w, \frac{\partial w}{\partial t}, \frac{\partial w}{\partial x_i} \in C(\overline{Q}), i = 1, \ldots, n \right\}$$

$$C^{1,2}(\overline{Q}) = \left\{ w(t,x) | w, \frac{\partial w}{\partial t}, \frac{\partial w}{\partial x_i}, \frac{\partial^2 w}{\partial x_i \partial x_j} \in C(\overline{Q}), i, j = 1, \ldots, n \right\}$$

$$C^{2,2}(\overline{Q}) = \left\{ w(t,x) | w, \frac{\partial w}{\partial t}, \frac{\partial^2 w}{\partial t^2}, \frac{\partial w}{\partial x_i}, \frac{\partial^2 w}{\partial x_i \partial x_j} \in C(\overline{Q}), i, j = 1, \ldots, n \right\}.$$

Let us consider a boundary value problem for the linear parabolic equation

$$Lz = \frac{\partial z}{\partial t} - \sum_{i,j=1}^{n} \frac{\partial}{\partial x_i} \left(a_{ij}(t,x) \frac{\partial z}{\partial x_j} \right) + \sum_{i=1}^{n} b_i(t,x) \frac{\partial z}{\partial x_i} + c(t,x) z = g \qquad \text{in } Q \tag{2.1}$$

$$\left(l_1(x) \frac{\partial z}{\partial \nu_{\mathcal{A}}} + l_2(x) z \right) \Big|_{\Sigma} = 0. \tag{2.2}$$

Here we set $\frac{\partial z}{\partial \nu_{\mathcal{A}}} = \sum_{i,j=1}^{n} a_{ij}(t,x) \nu_j(x) \frac{\partial z}{\partial x_i}$. In the above problem we assume

$$a_{ij} \in C^{2,2}(\overline{Q}) \qquad a_{ij} = a_{ji} \qquad b_i \in C^{1,1}(\overline{Q}) \; c \in C^{1,0}(\overline{Q}) \tag{2.3}$$

where $i, j = 1, \ldots, n$ and the uniform ellipticity: there exists $\mu > 0$ such that

$$a(t,x,\zeta,\zeta) \equiv \sum_{i,j=1}^{n} a_{ij}(t,x) \zeta_i \zeta_j \geqslant \mu |\zeta|^2 \qquad \forall \zeta \in \mathbb{R}^n, \; (t,x) \in Q. \tag{2.4}$$

We note that

$$\frac{\partial z}{\partial \nu_{\mathcal{A}}} = a(t,x,\nu,\nabla z).$$

Suppose that $l_1, l_2 \in C^2(\partial\Omega)$ and

$$\text{either } l_1(x) > 0, \forall x \in \partial\Omega \qquad \text{or} \qquad l_1(x) \equiv 0 \text{ and } l_2(x) \equiv 1. \tag{2.5}$$

For our Carleman estimate we need a weight function with special properties. The existence of such a function is proved in [5, 7].

Lemma 2.1. *Let ω_0 be an arbitrary fixed sub-domain of Ω such that $\overline{\omega_0} \subset \omega$. Then there exists a function $\psi \in C^2(\overline{\Omega})$ such that*

$$\psi(x) > 0 \qquad \forall x \in \Omega \qquad \psi|_{\partial\Omega} = 0 \qquad |\nabla\psi(x)| > 0 \qquad \forall x \in \overline{\Omega \setminus \omega_0}. \tag{2.6}$$

Example of ψ. Let $\Omega = \{x; |x| < 1\}$ and $\omega_0 = \{x; |x| < \frac{1}{2}\}$. Then $\psi(x) = 1 - |x|^2$ satisfies (2.6).

Let

$$\ell(t) = t(T - t). \tag{2.7}$$

We set

$$\varphi(t,x) = \frac{\mathrm{e}^{\lambda\psi(x)}}{\ell(t)} \tag{2.8}$$

and

$$\alpha(t,x) = \frac{\mathrm{e}^{\lambda\psi(x)} - \mathrm{e}^{2\lambda\|\psi\|_{C(\overline{\Omega})}}}{\ell(t)} \tag{2.9}$$

where $\lambda > 0$ and the function ψ is defined in lemma 2.1. Moreover, we set

$$\gamma = \sum_{i,j=1}^{n} \|a_{ij}\|_{C^{1,2}(\overline{Q})} + \sum_{i=1}^{n} \|b_i\|_{C^{0,1}(\overline{Q})} + \|c\|_{L^\infty(Q)} \qquad \tilde{\gamma} = \sum_{i,j=1}^{n} \|a_{ij}\|_{C^{1,2}(\overline{Q})}.$$

The following lemma, called a Carleman estimate, is proved in [5, 7].

Lemma 2.2. *Let (2.3)–(2.5) be fulfilled and the functions φ, α be defined by (2.6)–(2.9). Then there exists a number $\hat{\lambda} > 0$ such that for an arbitrary $\lambda \geqslant \hat{\lambda}$ we can choose $s_0(\lambda)$ satisfying: there exists a constant $C_1 > 0$ such that for each $s \geqslant s_0(\lambda)$, the solution $z \in W_2^{1,2}(Q)$ of the problem (2.1), (2.2) satisfies the following inequality:*

$$\int_Q \left((s\varphi)^{p-1} \left(\left| \frac{\partial z}{\partial t} \right|^2 + \sum_{i,j=1}^{n} \left| \frac{\partial^2 z}{\partial x_i \partial x_j} \right|^2 \right) + (s\varphi)^{p+1} |\nabla z|^2 + (s\varphi)^{p+3} z^2 \right) \mathrm{e}^{2s\alpha}\, \mathrm{d}x\, \mathrm{d}t$$
$$\leqslant C_1 \left(\int_Q (s\varphi)^p |g|^2 \mathrm{e}^{2s\alpha}\, \mathrm{d}x\, \mathrm{d}t + \int_{[0,T]\times\omega} (s\varphi)^{p+3} z^2 \mathrm{e}^{2s\alpha}\, \mathrm{d}x\, \mathrm{d}t \right) \qquad p = 0, 1 \tag{2.10}$$

where the constant $C_1 > 0$ depends continuously on γ, λ, but independent of s and the constant $\hat{\lambda}$ depends continuously on $\tilde{\gamma}$.

Remark. By (2.7)–(2.9), we note that

$$\lim_{t\downarrow 0} \varphi(t,x)^k \mathrm{e}^{2s\alpha(t,x)} = \lim_{t\uparrow T} \varphi(t,x)^k \mathrm{e}^{2s\alpha(t,x)} = 0$$

for any $k \geqslant 0$, and near $t = 0$ and $t = T$, the weight function $\mathrm{e}^{2s\alpha(t,x)}$ behaves with the same order of $\exp(-\frac{C}{t(T-t)})$ with $C > 0$. Therefore the norms in our Carleman estimate annihilate behaviour of functions at $t = 0$ and $t = T$.

For the case where we have overdeterminating data on a part of $\partial\Omega$ we need a weight function ψ with the following properties ([9]).

Lemma 2.3. *Let $\Gamma_0 \neq \emptyset$, $\subset \partial\Omega$ be an arbitrary relatively open set. Then there exists a function $\psi \in C^2(\overline{\Omega})$ such that*

$$\begin{aligned} &\psi(x) > 0 \qquad \forall x \in \Omega \qquad |\nabla\psi(x)| > 0 \qquad \forall x \in \overline{\Omega} \\ &\sum_{i,j=1}^{n} a_{ij}(t,x)\psi_{x_i}\nu_j \leqslant 0 \qquad (t,x) \in [0,T] \times (\partial\Omega \setminus \Gamma_0). \end{aligned} \tag{2.11}$$

Proof. Let us enlarge the domain Ω up to a domain $\tilde{\Omega}$ which has the following properties:

$$\Omega \subset \tilde{\Omega} \qquad \partial\Omega \setminus \Gamma_0 \subset \partial\tilde{\Omega} \qquad \mathrm{Int}(\tilde{\Omega} \setminus \Omega) \neq \emptyset.$$

Choose a sub-domain ω such that $\overline{\omega} \subset \mathrm{Int}(\tilde{\Omega} \setminus \Omega)$. Thus, by lemma 2.1 there exists a function $\psi(x)$ in $\tilde{\Omega}$ which satisfies $\psi(x) > 0$, $x \in \tilde{\Omega}$, $\psi_{|\partial\tilde{\Omega}} = 0$ and $|\nabla\psi(x)| > 0$, $x \in \overline{\tilde{\Omega} \setminus \omega}$. Therefore $\psi(x) > 0$, $x \in \Omega$ and $|\nabla\psi(x)| > 0$, $x \in \overline{\Omega}$. Finally, we have to verify $\sum_{i,j=1}^{n} a_{ij}(t,x)\psi_{x_i}\nu_j \leqslant 0$, $(t,x) \in [0,T] \times (\partial\Omega \setminus \Gamma_0)$. Since $\partial\Omega \setminus \Gamma_0 \subset \partial\tilde{\Omega}$, we

have $\psi_{|\partial\Omega\setminus\Gamma_0} = 0$ from $\psi_{|\partial\tilde{\Omega}} = 0$, which implies $\nabla\psi(x) = \frac{\partial\psi}{\partial\nu}(x)\nu(x)$, $x \in \partial\Omega \setminus \Gamma_0$, and so $\nabla\psi$ is parallel to ν on $\partial\Omega \setminus \Gamma_0$. Therefore $\nabla\psi(x) = -|\nabla\psi(x)|\nu(x)$, $x \in \partial\Omega \setminus \Gamma_0$. By $|\nabla\psi(x)| > 0$ on $\overline{\Omega}$, we obtain $\nu(x) = -\frac{\nabla\psi(x)}{|\nabla\psi(x)|}$, $x \in \partial\Omega \setminus \Gamma_0$. Therefore by (2.4) we see

$$\sum_{i,j=1}^{n} a_{ij}(t,x)\psi_{x_i}\nu_j = -\frac{\sum_{i,j=1}^{n} a_{ij}(t,x)\psi_{x_i}\psi_{x_j}}{|\nabla\psi(x)|} \leqslant 0 \qquad x \in \partial\Omega \setminus \Gamma_0.$$

□

By ψ constructed in lemma 2.3, we have the following.

Lemma 2.4. *Let (2.3)–(2.5) be fulfilled and the functions φ, α be defined by (2.11), (2.7)–(2.9). Then there exists a number $\hat{\lambda} > 0$ such that for an arbitrary $\lambda \geqslant \hat{\lambda}$ we can choose $s_0(\lambda)$ such that for each $s \geqslant s_0(\lambda)$, a solution $z \in W_2^{1,2}(Q)$ of the problem (2.1), (2.2) satisfies the following inequality:*

$$\begin{aligned}\int_Q \left((s\varphi)^{p-1}\left(\left|\frac{\partial z}{\partial t}\right|^2 + \sum_{i,j=1}^{n}\left|\frac{\partial^2 z}{\partial x_i \partial x_j}\right|^2\right) + (s\varphi)^{p+1}|\nabla z|^2 + (s\varphi)^{p+3}z^2\right)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \\ \leqslant C_1\int_Q (s\varphi)^p|g|^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \\ + C_1\int_{\Sigma_0}\left((s\varphi)^p\left(\frac{\partial z}{\partial t}\right)^2 + (s\varphi)^{p+1}|\nabla z|^2 + (s\varphi)^{p+3}z^2\right)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \qquad p=0,1\end{aligned} \tag{2.12}$$

where the constant C_1 depends continuously on γ, λ but independent of s, and the constant $\hat{\lambda}$ depends continuously on $\tilde{\gamma}$.

The proof of this lemma is similar to the proof of lemma 2.2 which is given in [5] or [7]. We can further refer to Imanuvilov [9].

3. Main theorems

We can now prove our main results on Lipschitz stability in our inverse problems. We recall that $Q =]0, T[\times\Omega$ and we can take $\delta > 0$ such that

$$0 \leqslant \theta - \delta < \theta < \theta + \delta \leqslant T. \tag{3.1}$$

We set

$$Q^{\omega} =]\theta - \delta, \theta + \delta[\times\omega. \tag{3.2}$$

Moreover we set

$$\begin{aligned}\varphi(t,x) = \varphi_\theta(t,x) &= \frac{\mathrm{e}^{\lambda\psi(x)}}{(t-(\theta-\delta))(\theta+\delta-t)} \\ \alpha(t,x) = \alpha_\theta(t,x) &= \frac{\mathrm{e}^{\lambda\psi(x)} - \mathrm{e}^{2\lambda\|\psi\|_{C(\overline{\Omega})}}}{(t-(\theta-\delta))(\theta+\delta-t)}\end{aligned} \tag{3.3}$$

for $(t,x) \in]\theta-\delta, \theta+\delta[\times\Omega$, where $\psi = \psi(x)$ is defined in lemma 2.1.

Theorem 3.1. *Let $g = g(t,x) \in W_2^{1,0}(Q)$ satisfy (1.2), $\theta \in]0, T[$ and conditions (2.3)–(2.5) be fulfilled. If $y \in W_2^{1,2}(Q)$ satisfies (1.1), then we have*

$$\|g\|_{L^2(Q)} \leqslant C(\|(y(\theta,\cdot)\|_{W_2^2(\Omega)} + \|\mathrm{e}^{s\alpha}y_t\|_{L^2(Q^\omega)} + \|\mathrm{e}^{s\alpha}y\|_{L^2(Q^\omega)}) \tag{3.4}$$

where the parameter $\hat{\lambda}$ is defined in lemma 2.2, $\lambda > \hat{\lambda}$, and $s > 0$ is sufficiently large.

Remark. We can uniquely determine and estimate $g = g(t, x)$ depending on both t and x in terms of single-measured data $y_{|]\theta-\delta,\theta+\delta[\times\omega}$, provided that g satisfies (1.2). In general, nonuniqueness occurs without (1.2) (see, e.g. [10, p 159]).

Proof. Since we can change scales of t, noting (1.2), without loss of generality, we can assume that $\theta = \frac{T}{2}$ and $\delta = \frac{T}{2}$ in (1.1). Then we note

$$\begin{aligned} \ell(t) &= t(T-t) \\ \varphi(t,x) &= \frac{\mathrm{e}^{\lambda\psi(x)}}{t(T-t)} \\ \alpha(t,x) &= \frac{\mathrm{e}^{\lambda\psi(x)} - \mathrm{e}^{2\lambda\|\psi\|_{C(\overline{\Omega})}}}{t(T-t)} \end{aligned} \tag{3.5}$$

for $(t, x) \in Q$, and $Q^\omega =]0, T[\times\omega$. By lemma 2.2 with $p = 1$, for a solution of the problem (1.1), we have the estimate

$$\begin{aligned} &\int_Q \left(\left(\left| \frac{\partial y}{\partial t} \right|^2 + \sum_{i,j=1}^n \left| \frac{\partial^2 y}{\partial x_i \partial x_j} \right|^2 \right) + s^2\varphi^2|\nabla y|^2 + s^4\varphi^4 y^2 \right) \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \\ &\qquad \leqslant C_1 \left(\int_{Q^\omega} s^4\varphi^4 y^2 \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t + \int_Q s\varphi g^2 \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \right) \qquad \forall s \geqslant s_0. \end{aligned} \tag{3.6}$$

By (1.1), the function $z(t, x) = \partial_t y(t, x)$ satisfies the system of equations:

$$z_t = -\mathcal{A}z + A_1 y + g_t \qquad \text{in } Q \tag{3.7}$$

$$\left(\ell_1 \frac{\partial z}{\partial \nu_{\mathcal{A}}} + \ell_2 z \right)|_{\partial\Omega} = 0 \tag{3.8}$$

$$z\left(\frac{T}{2}, \cdot\right) = -\mathcal{A}y_0 + g\left(\frac{T}{2}, \cdot\right) \tag{3.9}$$

where

$$A_1 y(t,x) = \sum_{i,j=1}^n \frac{\partial}{\partial x_i} \left(\frac{\partial a_{ij}(t,x)}{\partial t} \frac{\partial y}{\partial x_j}(t,x) \right) - \sum_{i=1}^n \frac{\partial b_i(t,x)}{\partial t} \frac{\partial y}{\partial x_i}(t,x) - \frac{\partial c(t,x)}{\partial t} y(t,x). \tag{3.10}$$

Since all the conditions of lemma 2.2 hold true in (3.7)–(3.9), setting $p = 0$ in that lemma and noting $|A_1 y(t,x)| \leqslant C_1 \sum_{|\beta|\leqslant 2} |(D_x^\beta y)(t,x)|$, we have the estimate

$$\begin{aligned} &\int_Q \left(\frac{1}{s\varphi} \left(\left| \frac{\partial z}{\partial t} \right|^2 + \sum_{i,j=1}^n \left| \frac{\partial^2 z}{\partial x_i \partial x_j} \right|^2 \right) + s\varphi|\nabla z|^2 + s^3\varphi^3 z^2 \right) \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \\ &\qquad \leqslant C_1 \left(\int_Q \left(\sum_{|\beta|\leqslant 2} |D_x^\beta y|^2 + |g_t|^2 \right) \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \right. \\ &\qquad\quad \left. + \int_{Q^\omega} s^3\varphi^3 z^2 \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \right) \qquad \forall s \geqslant s_0. \end{aligned} \tag{3.11}$$

Therefore we apply (3.6) for estimation of $|D_x^\beta y|$ in (3.11), increasing the parameter s if necessary, and we obtain

$$\int_Q \left(\frac{1}{s\varphi} \left(\left| \frac{\partial z}{\partial t} \right|^2 + \sum_{i,j=1}^n \left| \frac{\partial^2 z}{\partial x_i \partial x_j} \right|^2 \right) + s\varphi|\nabla z|^2 + s^3\varphi^3 z^2 \right) \mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$

$$\leqslant C_1\bigg(\int_{Q^\omega}(s^3\varphi^3z^2+s^4\varphi^4y^2)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$
$$+\int_Q(s\varphi g^2+|g_t|^2)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t\bigg)\qquad\forall s\geqslant s_0. \tag{3.12}$$

Henceforth $C>0$ denotes a generic constant which is independent of $(t,x)\in\overline{Q}$, y, z and a large parameter s. By (1.2) there exists a constant C such that

$$|g(t,x)|+|g_t(t,x)|\leqslant C\left|g\left(\frac{T}{2},x\right)\right|\qquad\forall(t,x)\in Q. \tag{3.13}$$

In fact, the mean value theorem implies $g(t,x)-g(\frac{T}{2},x)=g_t(\mu(t,x),x)(t-\frac{T}{2})$ for $(t,x)\in\overline{Q}$ where a number $\mu(t,x)$ is between t and $\frac{T}{2}$, and condition (1.2) implies $|g(t,x)|\leqslant TC|g(\frac{T}{2},x)|+|g(\frac{T}{2},x)|$ for almost all $(t,x)\in Q$.

Then (3.12) and (3.13) immediately yield

$$I_1\equiv\int_Q\left(\frac{1}{s\varphi}\left(\left|\frac{\partial z}{\partial t}\right|^2+\sum_{i,j=1}^n\left|\frac{\partial^2 z}{\partial x_i\partial x_j}\right|^2\right)+s\varphi|\nabla z|^2+s^3\varphi^3z^2\right)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$
$$\leqslant C_1\bigg(\int_{Q^\omega}(s^3\varphi^3y_t^2+s^4\varphi^4y^2)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$
$$+\int_Q s\varphi g\left(\frac{T}{2},x\right)^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t\bigg)\qquad\forall s\geqslant s_0. \tag{3.14}$$

On the other hand, in terms of $z^2(0,x)\mathrm{e}^{2s\alpha(0,x)}=0$ by (3.5) and $|\alpha_t|\leqslant C\varphi^2$ on $\overline{Q}$, we obtain

$$\int_\Omega sz^2\left(\frac{T}{2},x\right)\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x=\int_0^{\frac{T}{2}}\left(\frac{\partial}{\partial t}\int_\Omega sz^2(t,x)\mathrm{e}^{2s\alpha(t,x)}\,\mathrm{d}x\right)\mathrm{d}t$$
$$=\int_{]0,\frac{T}{2}[\times\Omega}\left(2s^2\frac{\partial\alpha}{\partial t}z^2+2sz\frac{\partial z}{\partial t}\right)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$
$$\leqslant C\int_Q\left\{s^2\varphi^2z^2\mathrm{e}^{2s\alpha}+2\left(\frac{1}{\sqrt{s\varphi}}\frac{\partial z}{\partial t}\mathrm{e}^{s\alpha}\right)\left(s\sqrt{s\varphi}z\mathrm{e}^{s\alpha}\right)\right\}\,\mathrm{d}x\,\mathrm{d}t$$
$$\leqslant C\int_Q s^2\varphi^2z^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t+C\int_Q\frac{1}{s\varphi}\left|\frac{\partial z}{\partial t}\right|^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$
$$+C\int_Q s^3\varphi z^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t\leqslant CI_1\qquad\forall s\geqslant s_0. \tag{3.15}$$

Combining (3.15) and (3.9), we obtain

$$\int_\Omega sg\left(\frac{T}{2},x\right)^2\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x\leqslant C\bigg(I_1+\int_\Omega s\sum_{|\beta|\leqslant 2}|D_x^\beta y_0|^2\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x\bigg). \tag{3.16}$$

Further, for large $s>0$, we can prove that there exists a constant $C>0$ such that

$$\int_Q s\varphi(t,x)g\left(\frac{T}{2},x\right)^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t\leqslant C\sqrt{s}\int_\Omega g\left(\frac{T}{2},x\right)^2\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x\qquad\forall s\geqslant s_1. \tag{3.17}$$

In fact, by (3.5) we have $\frac{\partial\alpha}{\partial t}(\frac{T}{2},x)=0$, $x\in\Omega$ and

$$\frac{\partial^2\alpha}{\partial t^2}(t,x)=\frac{2\ell_t(t)^2-\ell(t)\ell_{tt}(t)}{\ell(t)^3}(\mathrm{e}^{\lambda\psi(x)}-\mathrm{e}^{2\lambda\|\psi\|_{C(\overline{\Omega})}})$$

$$\frac{\partial^3\alpha}{\partial t^3}(t,x) = \frac{6(\ell(t)\ell_{tt}(t) - \ell_t(t)^2)}{\ell(t)^4}\ell_t(t)(\mathrm{e}^{\lambda\psi(x)} - \mathrm{e}^{2\lambda\|\psi\|_{C(\overline{\Omega})}})$$

for $(t,x) \in Q$, so that

$$\frac{\partial^2\alpha}{\partial t^2}(t,x) \leqslant -\frac{\mu_0}{\ell(t)^3} \qquad (t,x) \in \overline{Q}$$

with a positive constant μ_0, and

$$\frac{\partial^3\alpha}{\partial t^3}(t,x) \geqslant 0 \qquad 0 \leqslant t \leqslant \frac{T}{2} \qquad x \in \Omega$$
$$\frac{\partial^3\alpha}{\partial t^3}(t,x) \leqslant 0 \qquad \frac{T}{2} \leqslant t \leqslant T \qquad x \in \Omega$$

by (3.5). Therefore by the mean value theorem, we can take a number $\kappa = \kappa(t,x)$, $(t,x) \in Q$, such that κ is between t and $\frac{T}{2}$ and

$$\begin{aligned}\alpha(t,x) &= \alpha\left(\frac{T}{2},x\right) + \frac{1}{2}\frac{\partial^2\alpha}{\partial t^2}(t,x)\left(t-\frac{T}{2}\right)^2 + \frac{1}{6}\frac{\partial^3\alpha}{\partial t^3}(\kappa,x)\left(t-\frac{T}{2}\right)^3 \\ &\leqslant \alpha\left(\frac{T}{2},x\right) - \frac{\mu_0}{2t^3(T-t)^3}\left(t-\frac{T}{2}\right)^2 \qquad (t,x) \in Q.\end{aligned}$$

Hence, noting $\mu_0 > 0$ and $-\frac{1}{t(T-t)} \leqslant -\frac{4}{T^2}$, we obtain

$$\begin{aligned}\int_0^T \varphi(t,x)\mathrm{e}^{2s\alpha(t,x)}\,\mathrm{d}t &\leqslant C\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\int_0^T \frac{1}{t(T-t)}\exp\left(-\frac{s\mu_0}{t^3(T-t)^3}\left(t-\frac{T}{2}\right)^2\right)\mathrm{d}t \\ &\leqslant C\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\int_0^T \frac{1}{t(T-t)} \\ &\quad\times \exp\left(-\frac{\mu_0}{t^3(T-t)^3}\left(t-\frac{T}{2}\right)^2\right)\exp\left(-\frac{(s-1)\mu_0}{t^3(T-t)^3}\left(t-\frac{T}{2}\right)^2\right)\mathrm{d}t \\ &\leqslant C\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\int_0^T \exp\left(-\left(\frac{4}{T^2}\right)^3\mu_0(s-1)\left(t-\frac{T}{2}\right)^2\right)\mathrm{d}t \\ &\leqslant C\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\int_{-\infty}^{\infty}\exp\left(-\left(\frac{4}{T^2}\right)^3\mu_0(s-1)\eta^2\right)\mathrm{d}\eta \\ &= C\left(\frac{T^2}{4}\right)^{\frac{3}{2}}\sqrt{\frac{\pi}{\mu_0}}\frac{1}{\sqrt{s-1}}\mathrm{e}^{2s\alpha(\frac{T}{2},x)},\end{aligned}$$

which proves (3.17).

By (3.14), (3.17) we obtain from (3.16)

$$\begin{aligned}\int_\Omega sg\left(\frac{T}{2},x\right)^2\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x &\leqslant C\int_{Q^\omega}(s^3\varphi^3y_t^2 + s^4\varphi^4y^2)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t \\ &\quad + C\sqrt{s}\int_\Omega g\left(\frac{T}{2},x\right)^2\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x + Cs\int_\Omega\sum_{|\beta|\leqslant 2}|D_x^\beta y_0|^2\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x\end{aligned}$$

and for large $s > 0$, we obtain

$$\int_\Omega sg^2\left(\frac{T}{2},x\right)\mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x \leqslant C\int_{Q^\omega}(s^3\varphi^3y_t^2 + s^4\varphi^4y^2)\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t$$

$$+Cs\int_\Omega \sum_{|\beta|\leqslant 2} |D_x^\beta y_0|^2 \mathrm{e}^{2s\alpha(\frac{T}{2},x)}\,\mathrm{d}x.$$

From this inequality and (3.13), we obtain (3.4). □

For θ satisfying (3.1), we set $\Sigma_0 =]\theta-\delta,\theta+\delta[\times\Gamma_0$. Now let us consider the case of overdetermination on the part of the boundary.

Theorem 3.2. *Let $g = g(t,x) \in W_2^{1,0}(Q)$ satisfy (1.2), $\theta \in]0,T[$ and the conditions (2.3)–(2.5) be fulfilled. If $y \in W_2^{1,2}(Q)$ satisfies (1.3), then*

$$\|g\|_{L^2(Q)} \leqslant C(\|(y(\theta,\cdot)\|_{W_2^2(\Omega)} + \|y_t\mathrm{e}^{s\alpha}\|_{L^2(\Sigma_0)} + \|y\mathrm{e}^{s\alpha}\|_{L^2(\Sigma_0)} + \|\mathrm{e}^{s\alpha}\nabla y_t\|_{L^2(\Sigma_0)} + \|\mathrm{e}^{s\alpha}\nabla y\|_{L^2(\Sigma_0)}) \tag{3.18}$$

where the function α is defined by means of (3.3) with $\psi = \psi(x)$ given in lemma 2.3, $\lambda > \hat{\lambda}$, $s > 0$ is sufficiently large and the parameter $\hat{\lambda}$ is defined in lemma 2.4.

The proof of theorem 3.2 is very similar to the proof of theorem 3.1. In fact, only one difference is that one should use lemma 2.4 instead of lemma 2.2.

For inverse problems for (1.5) and (1.6), from theorems 3.1 and 3.2, we can directly prove the stability.

Theorem 3.3. *Let $R = R(t,x) \in C^{1,0}(\overline{Q})$ satisfy (1.4), $\theta \in]0,T[$ and conditions (2.3)–(2.5) be fulfilled. If $y \in W_2^{1,2}(Q)$ satisfies (1.5), then we have*

$$\|f\|_{L^2(\Omega)} \leqslant C(\|(y(\theta,\cdot)\|_{W_2^2(\Omega)} + \|y_t\mathrm{e}^{s\alpha}\|_{L^2(Q^\omega)} + \|y\mathrm{e}^{s\alpha}\|_{L^2(Q^\omega)}) \tag{3.19}$$

where the function α is defined by (3.5), the parameter $\hat{\lambda}$ is defined in lemma 2.2, $\lambda > \hat{\lambda}$, and $s > 0$ is sufficiently large.

Theorem 3.4. *Let $R = R(t,x) \in C^{1,0}(\overline{Q})$ satisfy (1.4), $\theta \in]0,T[$ and conditions (2.3)–(2.5) be fulfilled. If $y \in W_2^{1,2}(Q)$ satisfies (1.6), then*

$$\|f\|_{L^2(\Omega)} \leqslant C(\|(y(\theta,\cdot)\|_{W_2^2(\Omega)} + \|y_t\mathrm{e}^{s\alpha}\|_{L^2(\Sigma_0)} + \|y\mathrm{e}^{s\alpha}\|_{L^2(\Sigma_0)} + \|\mathrm{e}^{s\alpha}\nabla y_t\|_{L^2(\Sigma_0)} + \|\mathrm{e}^{s\alpha}\nabla y\|_{L^2(\Sigma_0)}) \tag{3.20}$$

where the function α is defined by means of (3.3) with $\psi = \psi(x)$ given in lemma 2.3, $\lambda > \hat{\lambda}$, $s > 0$ is sufficiently large and the parameter $\hat{\lambda}$ is defined in lemma 2.4.

4. Estimation of observation data by source term

In this section, we consider (1.5) and for simplicity, we assume that $\theta = \frac{T}{2}$, $\delta = \frac{T}{2}$ in (3.1), and $-\mathcal{A}$ is symmetric whose coefficients are smooth and independent of t, $y_0(x) = 0$, $x \in \Omega$, $l_1(x) = 0$, $l_2(x) = 1$, $x \in \partial\Omega$. We define an operator A in $L^2(\Omega)$ by

$$(Au)(x) = (\mathcal{A}u)(x) \qquad x \in \Omega \qquad \mathcal{D}(A) = \{u \in W_2^2(\Omega) | u_{|\partial\Omega} = 0\}.$$

Then it is known (see, e.g. [20]) that e^{-tA}, $t > 0$ is an analytic semigroup in $L^2(\Omega)$, and A has a compact resolvent.

Theorem 3.3 asserts that the norm defined by the right-hand side of (3.19) gives the unconditional Lipschitz continuity of the map: $y_{|]0,T[\times\omega} \longmapsto f \in L^2(\Omega)$. This result may be apparently contradictory, because the parabolic equation possesses the strong smoothing property (see e.g. [6, 20]). More precisely, in the case of $\theta = 0$, the map $f \longmapsto y$ can be proved to be a compact operator from $L^2(\Omega)$ to $W_2^{1,0}(Q)$, if $R \in C^{1,0}(\overline{Q})$.

In fact, we can assume that 0 is in the resolvent set of A, namely, A^{-1} exists and is a bounded operator. Then we can rewrite (1.5) as an evolution equation in $L^2(\Omega)$ (see, e.g. [20]), and we see

$$(A^{-1}y)_{tt} = -y_t + A^{-1}(fR_t) \qquad \text{in } Q$$
$$y(0,\cdot) = 0 \qquad \text{in } \Omega.$$

Since

$$y(t,\cdot) = \int_0^t \mathrm{e}^{-(t-\eta)A} fR(\eta,\cdot)\,\mathrm{d}\eta = \int_0^t \mathrm{e}^{-\eta A} fR(t-\eta,\cdot)\,\mathrm{d}\eta \qquad \text{in } L^2(\Omega)$$

we obtain

$$y_t(t,\cdot) = \mathrm{e}^{-tA} fR(0,\cdot) + \int_0^t \mathrm{e}^{-\eta A} fR_t(t-\eta,\cdot)\,\mathrm{d}\eta$$

and

$$\|y_t\|_{L^2(0,T;L^2(\Omega))} \leqslant C\|f\|_{L^2(\Omega)}.$$

Therefore

$$\|A^{-1}y_{tt}\|_{L^2(0,T;L^2(\Omega))} \leqslant C\|f\|_{L^2(\Omega)}. \tag{4.1}$$

On the other hand, A^κ, $\kappa \in \mathbb{R}$, can be defined, and for $\kappa > 0$ there exists a constant $C > 0$ such that

$$\|A^\kappa \mathrm{e}^{-\eta A}\| \leqslant C\eta^{-\kappa} \qquad \eta > 0$$

and $A^{-\kappa}$ is compact from $L^2(\Omega)$ to itself if $\kappa > 0$ (see, e.g. [20]). Hence, for fixed $\kappa \in]0,1[$, we can verify that

$$\|A^\kappa y_t\|_{L^2(0,T;L^2(\Omega))} \leqslant C\|f\|_{L^2(\Omega)}. \tag{4.2}$$

In view of (4.1) and (4.2), we can apply a compactness theorem (see, e.g. theorem III.2.1 (p 271) in [23]), so that the compactness of the map follows.

Hence, the inverse map $y_{|]0,T[\times\omega} \longmapsto f$ cannot be continuous with the topologies in $L^2(\Omega)$ and $W_2^{1,0}(Q)$. However, in system (1.5) with $\theta > 0$, the map $f \longmapsto y_{|]0,T[\times\omega}$ is not continuous from $L^2(\Omega)$ to $W_2^{1,0}(]0,T[\times\omega)$ because $f \longmapsto y_{|]0,\theta[\times\omega}$ involves the parabolic equation backward in t, which can account for our unconditional Lipschitz stability in the case of $\theta > 0$. In other words, in our formulation (1.5), the norm

$$\left(\int_{]0,T[\times\omega} (y(t,x)^2 + y_t(t,x)^2)\mathrm{e}^{2s\alpha(t,x)}\,\mathrm{d}x\,\mathrm{d}t\right)^{1/2}$$

is too strong in the sense that it cannot be, in general, estimated from the upper by $\|f\|_{L^2(\Omega)}$. For the Lipschitz stability in determining initial data with such 'too strong' norm, we can refer to Saitoh and Yamamoto [22], for example.

However, our choice of the norm of $y_{|]0,T[\times\omega}$ is not extremely strong. In fact, the upper estimation is true under some reasonable *a priori* boundedness assumption. That is, we prove the following.

Theorem 4.1. *We assume*

$$R \in C^{1,0}(\overline{Q}).$$

For an arbitrarily fixed $M > 0$, we define

$$\mathcal{F}_M = \left\{ f \in L^2(\Omega) \middle| \exists a \in L^2(\Omega) \text{ such that } \|a\|_{L^2(\Omega)} \leqslant M \right.$$
$$\left. \text{and } \mathrm{e}^{-\frac{T}{2}A} a + \int_0^{\frac{T}{2}} \mathrm{e}^{-(\frac{T}{2}-\eta)A} R(\eta, \cdot) f \, \mathrm{d}\eta = 0 \right\}.$$

Then for $f \in \mathcal{F}_M$ there exists a unique $y = y(f) \in C(]0, T]; W_2^2(\Omega)) \cap L^2(]0, T[\times\Omega)$ to

$$\begin{aligned} &y_t(t, x) = -\mathcal{A}y(t, x) + f(x)R(t, x) \qquad 0 < t < T, x \in \Omega \\ &y(t, x) = 0 \qquad 0 < t < T, x \in \partial\Omega \\ &y\left(\frac{T}{2}, x\right) = 0 \qquad x \in \Omega. \end{aligned} \tag{4.3}$$

Moreover for fixed $s > 0$, there exists a constant $C = C(M) > 0$ such that for all $f \in \mathcal{F}_M$,

$$\left(\int_{]0,T[\times\omega} \left(y(f)(t, x)^2 + \left| \frac{\partial y}{\partial t}(t, x) \right|^2 \right) \mathrm{e}^{2s\alpha(t,x)} \, \mathrm{d}x \, \mathrm{d}t \right)^{1/2}$$
$$\leqslant C_M \exp\left(-C_M \left| \log \frac{1}{\|f\|_{L^2(\Omega)}} \right|^{1/2} \right) \tag{4.4}$$

for $f \in \mathcal{F}_M$.

Remark. The stability rate in (4.4) is worse than any rate of Hölder type and better than a logarithmic rate. That is, for any $\beta > 0$, $x^\beta \exp(C_M(\log \frac{1}{x})^{1/2}) = \mathrm{o}(1)$ and $(\log \frac{1}{x})^\beta \exp(-C_M(\log \frac{1}{x})^{1/2}) = o(1)$ as $x \downarrow 0$. The *a priori* condition $f \in \mathcal{F}_M$ means that the $L^2(]0, T[\times\Omega)$ norm of our solution is uniformly bounded. When we consider some heat process, the *a priori* bound $M > 0$ for initial data is related for example with the melting point and so this type of *a priori* assumption may be physically realistic. The stability in the theorem holds true under such *a priori* information, and is called conditional stability. For other results on conditional stability for heat equations, we can refer to [3, 17, 19] and references in [4].

Proof. Let $f \in \mathcal{F}_M$. Then there exists a unique $a \in L^2(\Omega)$ satisfying

$$\mathrm{e}^{-\frac{T}{2}A} a + \int_0^{\frac{T}{2}} \mathrm{e}^{-(\frac{T}{2}-\eta)A} R(\eta, \cdot) f \, \mathrm{d}\eta = 0. \tag{4.5}$$

In fact, the existence of a follows from the definition of $\mathcal{F}_M$. Next we have to prove the uniqueness of such a. Let $\tilde{a} \in L^2(\Omega)$ satisfy (4.5). Then

$$\mathrm{e}^{-\frac{T}{2}A} a = \mathrm{e}^{-\frac{T}{2}A} \tilde{a}. \tag{4.6}$$

We know that we can take an orthonormal basis $\{\phi_n\}_{n\geqslant 1}$ of eigenfunctions of A:

$$A\phi_n = -\lambda_n \phi_n \qquad n \geqslant 1 \qquad (\phi_m, \phi_n) \equiv \int_\Omega \phi_m \phi_n \, \mathrm{d}x = \begin{cases} 1 & n = m \\ 0 & n \neq m \end{cases}.$$

Here we arrange all the eigenvalues of A as

$$\lambda_1 \leqslant \lambda_2 \leqslant \cdots\cdots \leqslant \lambda_n \leqslant \cdots \longrightarrow \infty$$

repeatedly according to their multiplicities. Equality (4.6) implies

$$\sum_{n=1}^{\infty} \mathrm{e}^{-\lambda_n \frac{T}{2}} (a, \phi_n)\phi_n = \sum_{n=1}^{\infty} \mathrm{e}^{-\lambda_n \frac{T}{2}} (\tilde{a}, \phi_n)\phi_n.$$

By the orthonormality of $\{\phi_n\}_{n\geqslant 1}$, we see that $(a, \phi_n) = (\tilde{a}, \phi_n)$, $n \geqslant 1$, namely, $a = \tilde{a}$ in $L^2(\Omega)$. Thus the proof of the uniqueness of a is complete.

Henceforth by $a = a(f)$ we denote $a \in L^2(\Omega)$ satisfying (4.5) for $f \in \mathcal{F}_M$. The solution $y = y(f)$ satisfies

$$\begin{aligned} &y_t(t, x) = -\mathcal{A}y(t, x) + f(x)R(t, x) \qquad 0 < t < T, x \in \Omega \\ &y(t, x) = 0 \qquad 0 < t < T, x \in \partial\Omega \\ &y(0, x) = a(f)(x) \qquad x \in \Omega. \end{aligned} \tag{4.7}$$

The regularity $y(f) \in C(]0, T]; W_2^2(\Omega)) \cap L^2(]0, T[\times\Omega)$ follows from a usual property of parabolic evolution equations (see, e.g. [20]).

For $f \in \mathcal{F}_M$, by (4.5), we have

$$\|\mathrm{e}^{-\frac{T}{2}A} a(f)\|_{L^2(\Omega)} = \left\| \int_0^{\frac{T}{2}} \mathrm{e}^{-(\frac{T}{2}-\eta)A} R(\eta, \cdot) f \,\mathrm{d}\eta \right\|_{L^2(\Omega)} \leqslant C\|f\|_{L^2(\Omega)}. \tag{4.8}$$

Hereafter $C > 0$ denotes a generic constant which is independent of f and dependent on $M > 0$, $\omega \subset \Omega$, $s > 0$, $T > 0$ and α. By $\|a(f)\|_{L^2(\Omega)} \leqslant M$, we apply the logarithmic convexity (see, e.g. [1, 19]), so that

$$\|\mathrm{e}^{-tA} a(f)\|_{L^2(\Omega)} \leqslant C\|\mathrm{e}^{-\frac{T}{2}A} a(f)\|_{L^2(\Omega)}^{\frac{2t}{T}} \qquad 0 \leqslant t \leqslant \frac{T}{2}.$$

Therefore by (4.8) we obtain

$$\|\mathrm{e}^{-tA} a(f)\|_{L^2(\Omega)} \leqslant C\|f\|_{L^2(\Omega)}^{\frac{2t}{T}} \qquad 0 \leqslant t \leqslant \frac{T}{2}. \tag{4.9}$$

By the property of ψ and ϕ, α, we have

$$|\mathrm{e}^{2s\alpha(t,x)}| \leqslant C \exp\left(-\frac{C}{t}\right) \qquad 0 < t < T, x \in \omega. \tag{4.10}$$

Therefore in view of (4.9), we obtain

$$\int_{]0,\frac{T}{2}[\times\omega} (\mathrm{e}^{-tA} a(f))(t, x)^2 \mathrm{e}^{2s\alpha(t,x)} \,\mathrm{d}x\,\mathrm{d}t \leqslant C \int_0^{\frac{T}{2}} \|f\|_{L^2(\Omega)}^{\frac{4t}{T}} \exp\left(-\frac{C}{t}\right) \mathrm{d}t.$$

Without loss of generality, we may assume that $\|f\|_{L^2(\Omega)} < 1$. We can then verify

$$\max_{t\geqslant 0} \|f\|_{L^2(\Omega)}^{\frac{4t}{T}} \exp\left(-\frac{C}{t}\right) \leqslant C \exp\left(-C \left|\log \frac{1}{\|f\|_{L^2(\Omega)}}\right|^{1/2}\right)$$

by direct calculations. Therefore

$$\int_{]0,\frac{T}{2}[\times\omega} (\mathrm{e}^{-tA} a(f))(t, x)^2 \mathrm{e}^{2s\alpha(t,x)} \,\mathrm{d}x\,\mathrm{d}t \leqslant C \exp\left(-C \left|\log \frac{1}{\|f\|_{L^2(\Omega)}}\right|^{1/2}\right). \tag{4.11}$$

On the other hand, by $R \in C^{1,0}(\overline{Q})$ and (4.8) we immediately see

$$\begin{aligned} \int_{]\frac{T}{2},T[\times\omega} (\mathrm{e}^{-tA} a(f))(t, x)^2 \mathrm{e}^{2s\alpha(t,x)} \,\mathrm{d}x\,\mathrm{d}t &\leqslant \int_{\frac{T}{2}}^{T} \|\mathrm{e}^{-tA} a(f)\|_{L^2(\Omega)}^2 \,\mathrm{d}t \\ &= \int_0^{\frac{T}{2}} \|\mathrm{e}^{-tA}\mathrm{e}^{-\frac{T}{2}A} a(f)\|_{L^2(\Omega)}^2 \,\mathrm{d}t \leqslant C\|f\|_{L^2(\Omega)}^2 \end{aligned} \tag{4.12}$$

and

$$\max_{0\leqslant t\leqslant T}\left\|\int_0^t \mathrm{e}^{-(t-\eta)A}R(\eta,\cdot)f\,\mathrm{d}\eta\right\|_{L^2(\Omega)}\leqslant C\|f\|_{L^2(\Omega)}. \tag{4.13}$$

Since $y(f)(t)=\mathrm{e}^{-tA}a(f)+\int_0^t \mathrm{e}^{-(t-\eta)A}R(\eta,\cdot)f\,\mathrm{d}\eta$, $0\leqslant t\leqslant T$, we use (4.11)–(4.13), so that we see

$$\left(\int_{]0,T[\times\omega} y(f)(t,x)^2\mathrm{e}^{2s\alpha(t,x)}\,\mathrm{d}x\,\mathrm{d}t\right)^{1/2}\leqslant C\exp\left(-C\left|\log\frac{1}{\|f\|_{L^2(\Omega)}}\right|^{1/2}\right)+C\|f\|_{L^2(\Omega)}$$
$$\leqslant C\exp\left(-C\left|\log\frac{1}{\|f\|_{L^2(\Omega)}}\right|^{1/2}\right).$$

Next we have to estimate

$$\left(\int_{]0,T[\times\omega}\left|\frac{\partial y(f)}{\partial t}(t,x)\right|^2\mathrm{e}^{2s\alpha(t,x)}\,\mathrm{d}x\,\mathrm{d}t\right)^{1/2}.$$

By integration by parts, we see

$$\frac{\partial y(f)}{\partial t}(t,\cdot)=-A\mathrm{e}^{-tA}a(f)+A^{-1}R(t,\cdot)f-A^{-1}\mathrm{e}^{-tA}R(0,\cdot)f$$
$$-\int_0^t A^{-1}\mathrm{e}^{-(t-s)A}\frac{\partial R}{\partial s}(s,\cdot)f\,\mathrm{d}s\qquad 0<t<T.$$

Therefore by (4.10), we have

$$\left(\int_{]0,T[\times\omega}\left|\frac{\partial y(f)}{\partial t}(t,x)\right|^2\mathrm{e}^{2s\alpha(t,x)}\,\mathrm{d}x\,\mathrm{d}t\right)^{1/2}\leqslant\left(\int_0^T\int_\omega|A\mathrm{e}^{-tA}a(f)|^2\mathrm{e}^{2s\alpha}\,\mathrm{d}x\,\mathrm{d}t\right)^{1/2}$$
$$+\left(\int_0^T\|A^{-1}R(t,\cdot)f-A^{-1}\mathrm{e}^{-tA}R(0,\cdot)f\|^2_{L^2(\Omega)}\,\mathrm{d}t\right)^{1/2}$$
$$+\left(\int_0^T\left\|\int_0^t A^{-1}\mathrm{e}^{-(t-s)A}\frac{\partial R}{\partial s}(s,\cdot)f\,\mathrm{d}s\right\|^2_{L^2(\Omega)}\mathrm{d}t\right)^{1/2}\equiv S_1+S_2+S_3.$$

By $R\in C^{1,0}(\overline{Q})$, we easily obtain

$$S_2+S_3\leqslant C\|f\|_{L^2(\Omega)}.$$

Hence for completing the proof of theorem 4.1, it suffices to prove that S_1 is estimated by the right-hand side of (4.4). By (4.10) we have

$$S_1\leqslant C\left(\int_0^{\frac{T}{2}}\|A\mathrm{e}^{-tA}a(f)\|^2_{L^2(\Omega)}\mathrm{e}^{-\frac{C}{t}}\,\mathrm{d}t\right)^{1/2}+C\left(\int_{\frac{T}{2}}^T\|A\mathrm{e}^{-tA}a(f)\|^2_{L^2(\Omega)}\,\mathrm{d}t\right)^{1/2}. \tag{4.14}$$

Since $\|A\mathrm{e}^{-tA}\|\leqslant\frac{C}{t}$, $t>0$ (see, e.g. [20]), from (4.9) we see

$$\|A\mathrm{e}^{-tA}a(f)\|^2_{L^2(\Omega)}\leqslant\frac{C}{t^2}\|\mathrm{e}^{-tA}a(f)\|^2_{L^2(\Omega)}\leqslant\frac{C}{t^2}\|f\|^{\frac{4t}{T}}_{L^2(\Omega)}\qquad 0<t\leqslant\frac{T}{2}. \tag{4.15}$$

Consequently, similarly to (4.11), we see

$$\int_0^{\frac{T}{2}}\|A\mathrm{e}^{-tA}a(f)\|^2_{L^2(\Omega)}\mathrm{e}^{-\frac{C}{t}}\,\mathrm{d}t\leqslant C\int_0^{\frac{T}{2}}\frac{1}{t^2}\|f\|^{\frac{4t}{T}}_{L^2(\Omega)}\mathrm{e}^{-\frac{C}{t}}\,\mathrm{d}t$$
$$\leqslant C\sup_{t\geqslant 0}\left(\frac{1}{t^2}\mathrm{e}^{-\frac{C}{2t}}\right)\int_0^{\frac{T}{2}}\|f\|^{\frac{4t}{T}}_{L^2(\Omega)}\mathrm{e}^{-\frac{C}{2t}}\,\mathrm{d}t\leqslant C\exp\left(-C\left|\log\frac{1}{\|f\|_{L^2(\Omega)}}\right|^{1/2}\right). \tag{4.16}$$

Moreover, in view of (4.15) with $t = \frac{T}{2}$, we have

$$\int_{\frac{T}{2}}^{T} \|A\mathrm{e}^{-tA}a(f)\|^2_{L^2(\Omega)}\,\mathrm{d}t = \int_0^{\frac{T}{2}} \|\mathrm{e}^{-tA}A\mathrm{e}^{-\frac{T}{2}A}a(f)\|^2_{L^2(\Omega)}\,\mathrm{d}t$$
$$\leqslant C\int_0^{\frac{T}{2}} \|A\mathrm{e}^{-\frac{T}{2}A}a(f)\|^2_{L^2(\Omega)}\,\mathrm{d}t \leqslant C\|f\|^2_{L^2(\Omega)}. \tag{4.17}$$

Application of (4.16) and (4.17) in (4.14) yields

$$S_1 \leqslant C\exp\left(-C\left|\log\frac{1}{\|f\|_{L^2(\Omega)}}\right|^{1/2}\right) + C\|f\|^2_{L^2(\Omega)}.$$

Thus the proof of theorem 4.1 is complete. □

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